

# Milestone 3 Report

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## 1 Forward Velocity Kinematics

For this milestone, the task was to get the kinematics for both the velocity and acceleration. Starting with the forward velocity kinematics, the goal is to map the joint velocities in the joint space to the end-effector velocities in the task space. To achieve this, the previously derived equations for the end-effector position, expressed in terms of the joint variables  $q$ , were used. The closed form equations for the end-effector position can be seen in Table 1.

| End-Effector Position | End-Effector Forward Position Equations                                                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x_{e.e}$             | $0.1 \sin(q_1) \cos(q_5) + 0.127 \sin(q_1) - 0.1 \sin(q_5) \cos(q_1) \cos(q_2 + q_3 + q_4) + 0.1 \sin(q_2 + q_3 + q_4) \cos(q_1) - 0.425 \cos(q_1) \cos(q_2) - 0.392 \cos(q_1) \cos(q_2 + q_3)$  |
| $y_{e.e}$             | $-0.1 \sin(q_1) \sin(q_5) \cos(q_2 + q_3 + q_4) + 0.1 \sin(q_1) \sin(q_2 + q_3 + q_4) - 0.425 \sin(q_1) \cos(q_2) - 0.392 \sin(q_1) \cos(q_2 + q_3) - 0.1 \cos(q_1) \cos(q_5) - 0.127 \cos(q_1)$ |
| $z_{e.e}$             | $-0.425 \sin(q_2) - 0.1 \sin(q_5) \sin(q_2 + q_3 + q_4) - 0.392 \sin(q_2 + q_3) - 0.1 \cos(q_2 + q_3 + q_4) + 0.163$                                                                             |

Table 1: Forward kinematic solution for the UR5E robot

Velocity is defined as the rate of change of position with respect to time. In other words, velocity is derived by differentiating the position with respect to time. However, since the position components are not functions of time, we use the chain rule:

$$\frac{dx}{dt} = \frac{dx}{dq} \cdot \frac{dq}{dt}$$

Since a robot arm has multiple joint variables  $q$ , the velocity of the end-effector's component can be expressed as partial derivatives. Each partial derivative is the velocity of a specific joint for a position component. Applying this to the UR5E robot arm, which has 6 joint variables  $q_1, q_2, q_3, q_4, q_5, q_6$ , results in the following end-effector's x-component velocity:

$$\frac{dx}{dt} = \frac{\partial x}{\partial q_1} \cdot \frac{dq_1}{dt} + \frac{\partial x}{\partial q_2} \cdot \frac{dq_2}{dt} + \frac{\partial x}{\partial q_3} \cdot \frac{dq_3}{dt} + \frac{\partial x}{\partial q_4} \cdot \frac{dq_4}{dt} + \frac{\partial x}{\partial q_5} \cdot \frac{dq_5}{dt} + \frac{\partial x}{\partial q_6} \cdot \frac{dq_6}{dt}$$

This relationship can be compactly represented in matrix form as:

$$\dot{X} = J \cdot \dot{q}$$

Here,  $\dot{X}$  is the end-effector velocities,  $\dot{q}$  is the joint velocities, and  $J$  is the Jacobian matrix, which contains the partial derivatives of the end-effector position with respect to each joint variable. It basically maps the joint velocities to the end-effector velocities. For the UR5E robot arm, expanding these matrices gives:

$$\begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{bmatrix} = \begin{bmatrix} J_{1,1} & J_{1,2} & J_{1,3} & J_{1,4} & J_{1,5} & J_{1,6} \\ J_{2,1} & J_{2,2} & J_{2,3} & J_{2,4} & J_{2,5} & J_{2,6} \\ J_{3,1} & J_{3,2} & J_{3,3} & J_{3,4} & J_{3,5} & J_{3,6} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \\ \dot{q}_5 \\ \dot{q}_6 \end{bmatrix}$$

Using the python library SymPy, each  $J_{i,j}$  was computed, with the resulting partial derivatives shown in Table 2.

| Jacobian Element                            | Partial Derivative (Equation)                                                                                                                                                                   |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $J_{1,1} = \frac{\partial x}{\partial q_1}$ | $0.1 \sin(q_1) \sin(q_5) \cos(q_2 + q_3 + q_4) - 0.1 \sin(q_1) \sin(q_2 + q_3 + q_4) + 0.425 \sin(q_1) \cos(q_2) + 0.392 \sin(q_1) \cos(q_2 + q_3) + 0.1 \cos(q_1) \cos(q_5) + 0.127 \cos(q_1)$ |
| $J_{1,2} = \frac{\partial x}{\partial q_2}$ | $0.425 \sin(q_2) \cos(q_1) + 0.1 \sin(q_5) \sin(q_2 + q_3 + q_4) \cos(q_1) + 0.392 \sin(q_2 + q_3) \cos(q_1) + 0.1 \cos(q_1) \cos(q_2 + q_3 + q_4)$                                             |
| $J_{1,3} = \frac{\partial x}{\partial q_3}$ | $0.1 \sin(q_5) \sin(q_2 + q_3 + q_4) \cos(q_1) + 0.392 \sin(q_2 + q_3) \cos(q_1) + 0.1 \cos(q_1) \cos(q_2 + q_3 + q_4)$                                                                         |
| $J_{1,4} = \frac{\partial x}{\partial q_4}$ | $0.1 \sin(q_5) \sin(q_2 + q_3 + q_4) \cos(q_1) + 0.1 \cos(q_1) \cos(q_2 + q_3 + q_4)$                                                                                                           |
| $J_{1,5} = \frac{\partial x}{\partial q_5}$ | $-0.1 \sin(q_1) \sin(q_5) - 0.1 \cos(q_1) \cos(q_5) \cos(q_2 + q_3 + q_4)$                                                                                                                      |
| $J_{1,6} = \frac{\partial x}{\partial q_6}$ | 0                                                                                                                                                                                               |
| $J_{2,1} = \frac{\partial y}{\partial q_1}$ | $0.1 \sin(q_1) \cos(q_5) + 0.127 \sin(q_1) - 0.1 \sin(q_5) \cos(q_1) \cos(q_2 + q_3 + q_4) + 0.1 \sin(q_2 + q_3 + q_4) \cos(q_1) - 0.425 \cos(q_1) \cos(q_2) - 0.392 \cos(q_1) \cos(q_2 + q_3)$ |
| $J_{2,2} = \frac{\partial y}{\partial q_2}$ | $0.425 \sin(q_1) \sin(q_2) + 0.1 \sin(q_1) \sin(q_5) \sin(q_2 + q_3 + q_4) + 0.392 \sin(q_1) \sin(q_2 + q_3) + 0.1 \sin(q_1) \cos(q_2 + q_3 + q_4)$                                             |
| $J_{2,3} = \frac{\partial y}{\partial q_3}$ | $0.1 \sin(q_1) \sin(q_5) \sin(q_2 + q_3 + q_4) + 0.392 \sin(q_1) \sin(q_2 + q_3) + 0.1 \sin(q_1) \cos(q_2 + q_3 + q_4)$                                                                         |
| $J_{2,4} = \frac{\partial y}{\partial q_4}$ | $0.1 \sin(q_1) \sin(q_5) \sin(q_2 + q_3 + q_4) + 0.1 \sin(q_1) \cos(q_2 + q_3 + q_4)$                                                                                                           |
| $J_{2,5} = \frac{\partial y}{\partial q_5}$ | $-0.1 \sin(q_1) \cos(q_5) \cos(q_2 + q_3 + q_4) + 0.1 \sin(q_5) \cos(q_1)$                                                                                                                      |
| $J_{2,6} = \frac{\partial y}{\partial q_6}$ | 0                                                                                                                                                                                               |
| $J_{3,1} = \frac{\partial z}{\partial q_1}$ | 0                                                                                                                                                                                               |
| $J_{3,2} = \frac{\partial z}{\partial q_2}$ | $-0.1 \sin(q_5) \cos(q_2 + q_3 + q_4) + 0.1 \sin(q_2 + q_3 + q_4) - 0.425 \cos(q_2) - 0.392 \cos(q_2 + q_3)$                                                                                    |
| $J_{3,3} = \frac{\partial z}{\partial q_3}$ | $-0.1 \sin(q_5) \cos(q_2 + q_3 + q_4) + 0.1 \sin(q_2 + q_3 + q_4) - 0.392 \cos(q_2 + q_3)$                                                                                                      |
| $J_{3,4} = \frac{\partial z}{\partial q_4}$ | $-0.1 \sin(q_5) \cos(q_2 + q_3 + q_4) + 0.1 \sin(q_2 + q_3 + q_4)$                                                                                                                              |
| $J_{3,5} = \frac{\partial z}{\partial q_5}$ | $-0.1 \sin(q_2 + q_3 + q_4) \cos(q_5)$                                                                                                                                                          |
| $J_{3,6} = \frac{\partial z}{\partial q_6}$ | 0                                                                                                                                                                                               |

Table 2: Jacobian elements  $J_{i,j}$  and their corresponding partial derivatives

## 2 Inverse Velocity Kinematics

The goal of inverse velocity kinematics is essentially the reverse of forward velocity kinematics. Instead of mapping joint velocities to end-effector velocities, it maps the end-effector velocities in task space to the corresponding joint velocities in joint space. This can be achieved by applying the inverse of the Jacobian matrix to both sides of the forward velocity equation:

$$\begin{aligned}\dot{X} &= J \cdot \dot{q} \\ \dot{q} &= J^{-1} \cdot \dot{X}\end{aligned}$$

However, since the Jacobian matrix derived for the UR5E robot arm is not a square matrix, the pseudo-inverse  $J^+$  is used instead, resulting in:

$$\dot{q} = J^+ \cdot \dot{X} = J^T (J J^T)^{-1} \cdot \dot{X}$$

Expanding the matrices results in the following:

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \\ \dot{q}_5 \\ \dot{q}_6 \end{bmatrix} = \begin{bmatrix} J_{1,1}^+ & J_{1,2}^+ & J_{1,3}^+ \\ J_{2,1}^+ & J_{2,2}^+ & J_{2,3}^+ \\ J_{3,1}^+ & J_{3,2}^+ & J_{3,3}^+ \\ J_{4,1}^+ & J_{4,2}^+ & J_{4,3}^+ \\ J_{5,1}^+ & J_{5,2}^+ & J_{5,3}^+ \\ J_{6,1}^+ & J_{6,2}^+ & J_{6,3}^+ \end{bmatrix} \begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{bmatrix}$$

Here, the Jacobian matrix shown is after applying the pseudo-inverse. Due to the intense computation of deriving explicit expressions for each  $J_{i,j}^+$ , their values are not shown and instead computed numerically once the joint values are known.

## 3 Forward Acceleration Kinematics

The forward acceleration kinematics builds on the forward velocity kinematics. To derive it, the forward velocity is differentiated with respect to time. Similarly to before, the chain rule is used. The product rule is also applied since they are two separate functions, giving us:

$$\begin{aligned}\ddot{X} &= \frac{d}{dt} \dot{X} = \frac{d}{dt} (J \dot{q}) \\ \ddot{X} &= \dot{J} \dot{q} + J \ddot{q}\end{aligned}$$

Here,  $\ddot{X}$  is the end-effector acceleration,  $\ddot{q}$  is the joint accelerations,  $\dot{q}$  is the joint velocities,  $J$  is the Jacobian matrix, and  $\dot{J}$  is its time derivative. This expression maps the joints accelerations to the end-effector acceleration. Expanding these matrices for the UR5E robot arm once again results in:

$$\begin{bmatrix} \ddot{X} \\ \ddot{Y} \\ \ddot{Z} \end{bmatrix} = \begin{bmatrix} \dot{J}_{1,1} & \dot{J}_{1,2} & \dot{J}_{1,3} & \dot{J}_{1,4} & \dot{J}_{1,5} & \dot{J}_{1,6} \\ \dot{J}_{2,1} & \dot{J}_{2,2} & \dot{J}_{2,3} & \dot{J}_{2,4} & \dot{J}_{2,5} & \dot{J}_{2,6} \\ \dot{J}_{3,1} & \dot{J}_{3,2} & \dot{J}_{3,3} & \dot{J}_{3,4} & \dot{J}_{3,5} & \dot{J}_{3,6} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \\ \dot{q}_5 \\ \dot{q}_6 \end{bmatrix} + \begin{bmatrix} J_{1,1} & J_{1,2} & J_{1,3} & J_{1,4} & J_{1,5} & J_{1,6} \\ J_{2,1} & J_{2,2} & J_{2,3} & J_{2,4} & J_{2,5} & J_{2,6} \\ J_{3,1} & J_{3,2} & J_{3,3} & J_{3,4} & J_{3,5} & J_{3,6} \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \\ \ddot{q}_3 \\ \ddot{q}_4 \\ \ddot{q}_5 \\ \ddot{q}_6 \end{bmatrix}$$

The values of  $J$  can be seen in Table 2. Due to the length of each element  $J_{i,j}$ , it is not shown in the report, however it can be seen by running the code `jacobian_dot_matrix.py` if needed.

## 4 Inverse Acceleration Kinematics

Inverse acceleration kinematics does the opposite of forward acceleration kinematics, where it maps the end-effector's acceleration to the joints accelerations. Similar to inverse velocity kinematics, this can be obtained through simple algebraic manipulation.

$$\ddot{X} = \dot{J}\dot{q} + J\ddot{q}$$

$$J\ddot{q} = \ddot{X} - \dot{J}\dot{q}$$

$$\boxed{\ddot{q} = J^+ \left( \ddot{X} - \dot{J}\dot{q} \right)}$$

As mentioned before,  $J^+$  is the psuedo-inverse of  $J$  due to it not being a square matrix. Expanding these matrices gives us:

$$\begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \\ \ddot{q}_3 \\ \ddot{q}_4 \\ \ddot{q}_5 \\ \ddot{q}_6 \end{bmatrix} = \begin{bmatrix} J_{1,1}^+ & J_{1,2}^+ & J_{1,3}^+ \\ J_{2,1}^+ & J_{2,2}^+ & J_{2,3}^+ \\ J_{3,1}^+ & J_{3,2}^+ & J_{3,3}^+ \\ J_{4,1}^+ & J_{4,2}^+ & J_{4,3}^+ \\ J_{5,1}^+ & J_{5,2}^+ & J_{5,3}^+ \\ J_{6,1}^+ & J_{6,2}^+ & J_{6,3}^+ \end{bmatrix} \left( \begin{bmatrix} \ddot{X} \\ \ddot{Y} \\ \ddot{Z} \end{bmatrix} - \begin{bmatrix} \dot{J}_{1,1} & \dot{J}_{1,2} & \dot{J}_{1,3} & \dot{J}_{1,4} & \dot{J}_{1,5} & \dot{J}_{1,6} \\ \dot{J}_{2,1} & \dot{J}_{2,2} & \dot{J}_{2,3} & \dot{J}_{2,4} & \dot{J}_{2,5} & \dot{J}_{2,6} \\ \dot{J}_{3,1} & \dot{J}_{3,2} & \dot{J}_{3,3} & \dot{J}_{3,4} & \dot{J}_{3,5} & \dot{J}_{3,6} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \\ \dot{q}_5 \\ \dot{q}_6 \end{bmatrix} \right)$$

All derived equations were computed using NumPy and SymPy built-in libraries.